WORKED SOLUTION

STEP I 1994, Question 2

Pure • Logarithmic Differentiation

> **STEP I 1994, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/94-s1-q2

The key idea

The power rule works when the variable is in the base and the exponent is fixed; the exponential rule works when the base is fixed and the variable is in the exponent. As soon as x appears in both places, neither rule applies directly. The cure is to write everything as a power of e, or equivalently to take logarithms and differentiate implicitly: logarithms turn awkward powers into products, and products we can handle.

Throughout, $x > 0$ (otherwise x^x and $\ln x$ make no sense), and in (ii) the base a must be positive.

Solution

The one tool we use repeatedly: for $u > 0$,

$$u^v = \mathrm{e}^{v \ln u},$$

so if $y = u^v$ then $\ln y = v \ln u$, and differentiating with the chain rule gives $\dfrac{1}{y}\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}}{\mathrm{d}x}(v \ln u)$.

Part (i)

Here the exponent is constant, so this is the ordinary power rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\, x^a = a\, x^{a-1}.$$

If you want to see why it holds for any real a (not just integers), write $x^a = \mathrm{e}^{a \ln x}$; the chain rule gives $\mathrm{e}^{a \ln x} \cdot \frac{a}{x} = a\, x^a / x = a\, x^{a-1}$.

Part (ii)

Now the base is constant. Writing $a^x = \mathrm{e}^{x \ln a}$, where $\ln a$ is just a constant,

$$\frac{\mathrm{d}}{\mathrm{d}x} a^x = \ln a \, \mathrm{e}^{x \ln a} = a^x \ln a.$$

Note how different this is from (i): it is *not* $x\, a^{x-1}$.

Part (iii)

Let $y = x^x$. Taking logarithms, $\ln y = x \ln x$. Differentiating both sides, using the product rule on the right,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x + x \cdot \frac{1}{x} = 1 + \ln x,$$

so

$$\frac{\mathrm{d}}{\mathrm{d}x} x^x = x^x \, (1 + \ln x).$$

? **Ask yourself** Why can't we just use the power rule and get $x \cdot x^{x-1} = x^x$?

The power rule assumes the exponent is constant, and the exponential rule assumes the base is constant. Here both vary. In fact the answer is exactly the sum of the two "wrong" answers: treating the exponent as fixed gives $x \cdot x^{x-1} = x^x$, treating the base as fixed gives $x^x \ln x$, and the true derivative is $x^x + x^x \ln x$. This is the multivariable chain rule in disguise: when $y = u^v$, each variable contributes its own rate of change.

Part (iv)

Let $y = x^{(x^x)}$, a tower in which the exponent is itself x^x. Taking logarithms,

$$\ln y = x^x \ln x.$$

Differentiate the right-hand side with the product rule, using (iii) for the derivative of x^x:

$$\begin{aligned}\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} &= x^x(1 + \ln x) \cdot \ln x + x^x \cdot \frac{1}{x} \\ &= x^x \left(\frac{1}{x} + \ln x + (\ln x)^2 \right).\end{aligned}$$

Multiplying by $y = x^{(x^x)}$ and combining the powers of x,

$$\frac{\mathrm{d}}{\mathrm{d}x} x^{(x^x)} = x^{(x^x)} \, x^x \left(\frac{1}{x} + \ln x + (\ln x)^2 \right) = x^{x^x + x} \left(\frac{1}{x} + \ln x + (\ln x)^2 \right).$$

Equivalently, taking the $\frac{1}{x}$ inside, $x^{x^x + x - 1}(1 + x \ln x + x(\ln x)^2)$.

Part (v)

? Ask yourself Isn't $(x^x)^x$ the same as the expression in (iv)?

No. By the law of indices $(u^m)^n = u^{mn}$, we have $(x^x)^x = x^{x\cdot x} = x^{x^2}$, whereas in (iv) the exponent is x^x, not x^2. They happen to agree at $x = 1$ and $x = 2$ (both give 16 at $x = 2$), but at $x = 3$ one is 3^{27} and the other only 3^9.

So let $y = x^{x^2}$. Then $\ln y = x^2 \ln x$, and by the product rule

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = 2x \ln x + x^2 \cdot \frac{1}{x} = x\,(1 + 2\ln x).$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\,(x^x)^x = x^{x^2} \cdot x\,(1 + 2\ln x) = x^{x^2+1}\,(1 + 2\ln x).$$

(You could instead differentiate $(x^x)^x$ directly: $\ln y = x\ln(x^x) = x^2 \ln x$, the same thing.)

Checks

- At $x = 1$ every one of (iii), (iv) and (v) gives derivative 1: for instance in (iv), $1^{1+1}(1 + 0 + 0) = 1$. A central difference $\big(f(1.000001) - f(0.999999)\big)/0.000002$ agrees.
- x^x has a minimum at $x = \mathrm{e}^{-1}$, where $1 + \ln x = 0$; and $(x^x)^x = x^{x^2}$ has a minimum at $x = \mathrm{e}^{-1/2}$, where $1 + 2\ln x = 0$. Both are visible on a graphical calculator.
- We compared every formula with a numerical derivative at 2000 random points with $0.2 \le x \le 2.5$ (and $0.2 \le a \le 4$); the relative error was always below 10^{-9}.

Common slips

- Writing $\frac{\mathrm{d}}{\mathrm{d}x}x^x = x \cdot x^{x-1}$ or $\frac{\mathrm{d}}{\mathrm{d}x}a^x = x\,a^{x-1}$: the power rule needs a constant exponent.
- Treating (iv) and (v) as the same function. Powers are not associative: $x^{(x^x)} \neq (x^x)^x$ in general.
- Forgetting to multiply back by y after differentiating $\ln y$.

Beyond STEP

What does x^a even mean when a is irrational, say $x^{\sqrt{2}}$? At university the answer is to *define* $u^v = \mathrm{e}^{v \ln u}$ for $u > 0$, after first defining exp by its power series and ln as its inverse; all the rules of indices and the derivatives above then become theorems. This is done carefully in Abbott, *Understanding Analysis*, or in Bartle and Sherbert, *Introduction to Real Analysis*. The "sum of the two wrong answers" in (iii) is a first glimpse of partial derivatives and the multivariable chain rule.

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